Chapter 4 – Probability Theory and Distributions

Mathematical Foundations for Machine Learning

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Learning Objectives

- Describe sample spaces, events, conditional probability and independence.
- Apply Bayes' theorem to update probabilities from evidence.
- Model discrete and continuous random variables using standard distributions.
- Compute expectations, variances and probabilities.
- Apply the Central Limit Theorem and continuity correction.
- Connect probabilistic models to machine learning and data analysis.

Outline

1. Introduction
2. Probability Spaces and Events
3. Conditional Probability and Independence
4. Bayes' Theorem
5. Discrete Random Variables
6. Continuous Random Variables
7. Central Limit Theorem and Continuity Correction
8. Conclusion

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Why Probability for Machine Learning?

- Observations contain uncertainty: spam emails, user clicks, sensor readings.
- Probability models unknown outcomes and quantifies uncertainty.
- Conditional probability updates predictions when evidence arrives.
- Probability distributions model labels, counts and continuous measurements.

Key idea: Probability provides the mathematical language of uncertainty used in statistical learning.

From Data to Probabilistic Prediction



Example: email filtering

A model estimates $P(\text{spam} \mid \text{observed words})$ and uses the probability to classify an email.

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Random Experiment and Sample Space

Definition

A random experiment has an outcome that cannot be predicted with certainty. The **sample space** Ω contains all possible outcomes.

Examples

- Coin toss: $\Omega = \{H, T\}$.
- Die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Temperature measurement: $\Omega = [-50, 60]$.

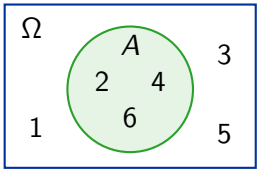
Events

Definition

An event is a subset $A \subseteq \Omega$. It occurs when the observed outcome belongs to A.

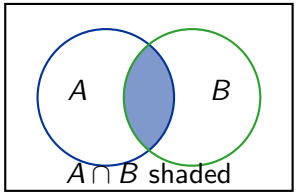
For a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$:

- $A = \{2, 4, 6\}$: even outcome.
- $B = \{4, 5, 6\}$: outcome > 3.
- Certain event: Ω.
- Impossible event: $\varnothing$.



Operations on Events

- Union $A \cup B$: A or B.
- Intersection $A \cap B$: both.
- Complement A^c: not A.
- Difference $A \setminus B$: A but not B.



For $A = \{2, 4, 6\}$, $B = \{4, 5, 6\}$:

$$A \cup B = \{2, 4, 5, 6\}, \quad A \cap B = \{4, 6\}.$$

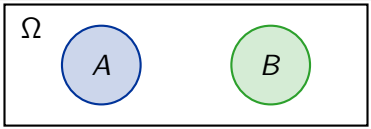
Union Rule and Disjoint Events

Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Disjoint events

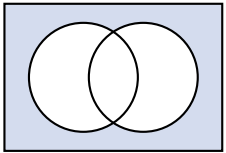
If $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.



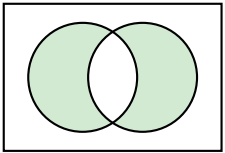
De Morgan's Laws

Two fundamental identities

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$



$(A \cup B)^c$



$A^c \cup B^c$ (concept)

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Conditional Probability

Definition

For $P(B) > 0$,

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}.$$

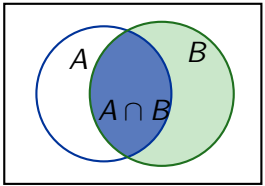
Key idea: Knowing B occurred restricts the relevant sample space to B.

Spam example

If $P(S) = .30$, $P(O) = .20$ and $P(S \cap O) = .15$, then

$$P(S \mid O) = \frac{.15}{.20} = .75.$$

Conditional Probability: Visual Interpretation



- Green region: conditioning event B.
- Blue overlap: outcomes in both A and B.
- Conditional probability = overlap probability divided by $P(B)$.

Independence

Definition

Events A and B are independent if

$$\boxed{P(A \cap B) = P(A)P(B)}.$$

If $P(B) > 0$, equivalently $P(A \mid B) = P(A)$.

Two fair coin tosses

$$P(H_1) = P(H_2) = \tfrac{1}{2}, \qquad P(H_1 \cap H_2) = \tfrac{1}{4}.$$

The two events are independent.

Dependence: Cards Without Replacement

Two draws from a standard deck

Let A be "first card is an ace" and B be "second card is an ace".

$$P(B \mid A) = \frac{3}{51}, \qquad P(B) = \frac{4}{52}.$$

Since these differ, the events are dependent.

Key idea: Sampling without replacement typically introduces dependence.

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Bayes' Theorem

Theorem

For $P(A) > 0$,

$$\boxed{P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}}.$$

$$\underbrace{P(B \mid A)}_{\text{Posterior}} = \frac{\overbrace{P(A \mid B)}^{\text{Likelihood}} \overbrace{P(B)}^{\text{Prior}}}{\underbrace{P(A)}_{\text{Evidence}}}$$

Bayesian Updating: Rain Prediction

Cloudy morning

Let $R =$ rain, $C =$ cloudy. Suppose

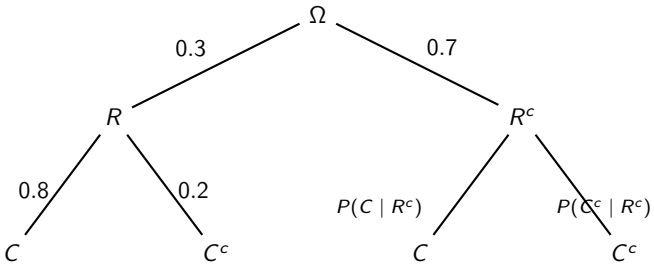
$$P(R) = .3, \quad P(C) = .4, \quad P(C \mid R) = .8.$$

Then

$$P(R \mid C) = \frac{.8 \times .3}{.4} = .6.$$

Key idea: The observed clouds update the probability of rain from 0.3 to 0.6.

Bayes' Theorem: Probability Tree



Joint Probability

$$P(R \cap C) = P(C \mid R)P(R) = 0.8 \times 0.3 = \boxed{0.24}.$$

- Classification estimates $P(\text{class} \mid \text{features})$.
- The prior captures class prevalence before observing features.
- The likelihood measures compatibility of features with each class.
- Naive Bayes additionally assumes conditional independence of features given the class.

Two-class evidence

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c).$$

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Discrete Random Variables

Definition

A discrete random variable maps outcomes to a finite or countable set of numerical values:

$$X : \Omega \longrightarrow \mathbb{R}, \qquad p_i = P(X = x_i).$$

Its probability mass function satisfies $p_i \geq 0$ and $\sum_i p_i = 1$.

Examples

Number of red balls drawn; number of defective items; number of website visits.

Expectation and Variance

Important formulas

$$\boxed{\mathbb{E}[X] = \sum_i x_i P(X = x_i)}$$

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2}$$

where $\mathbb{E}[X^2] = \sum_i x_i^2 P(X = x_i)$.

Key idea: Expectation locates the center; variance quantifies dispersion.

Worked Example: Drawing Balls

Urn with 2 red and 3 black balls

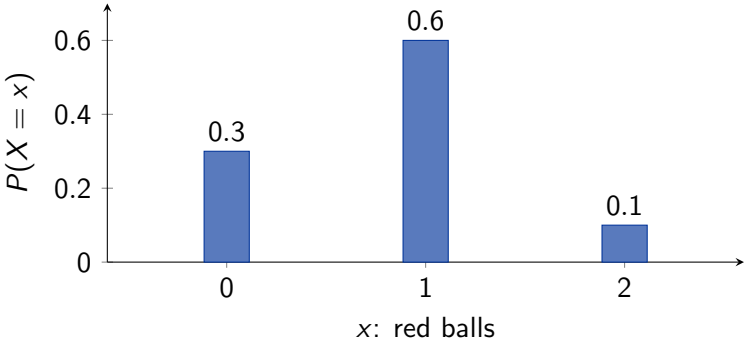
Draw two balls simultaneously. Let X be the number of red balls.

$$P(X=0)=\frac{3}{10}, \quad P(X=1)=\frac{6}{10}, \quad P(X=2)=\frac{1}{10}.$$

$$\mathbb{E}[X]=0(.3)+1(.6)+2(.1)=.8,$$

$$\mathbb{E}[X^2]=0+.6+4(.1)=1, \quad \mathrm{Var}(X)=1-.8^2=.36.$$

Urn Example: Probability Mass Function



Discrete Uniform Distribution

Definition

If X is equally likely to take each value in $\{1, \ldots, n\}$,

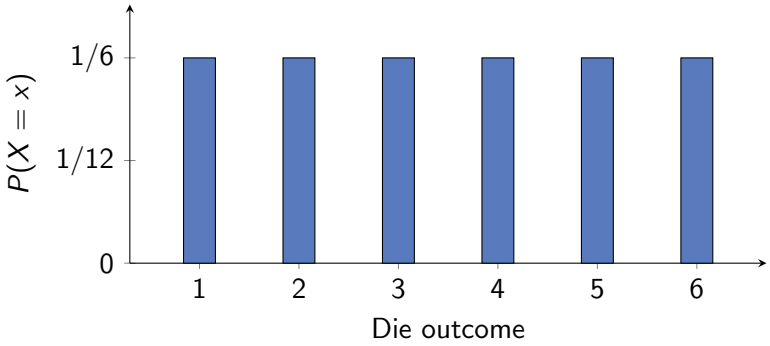
$$P(X = k) = \frac{1}{n}, \quad k = 1, \ldots, n.$$

$$\mathbb{E}[X] = \frac{n+1}{2}, \qquad \mathrm{Var}(X) = \frac{n^2 - 1}{12}.$$

Fair die

For $n = 6$, $\mathbb{E}[X] = 3.5$ and $\mathrm{Var}(X) = 35/12$.

Uniform Distribution: Fair Die



Bernoulli Distribution

Definition

For one trial with success probability p,

$$X = \begin{cases} 1 & \text{success}, \\ 0 & \text{failure}, \end{cases} \qquad X \sim \text{Bernoulli}(p).$$

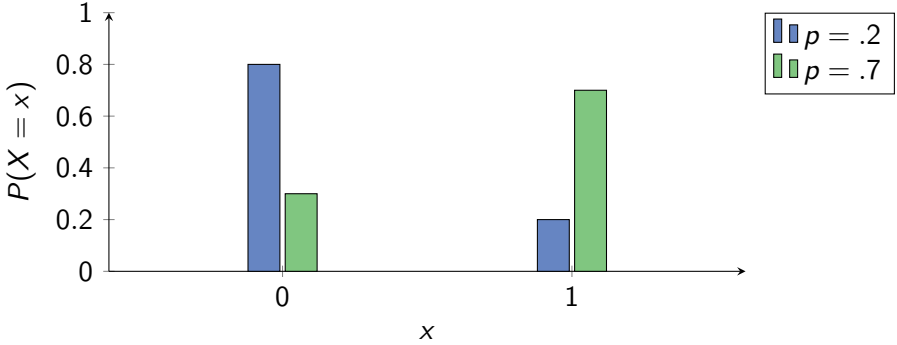
$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

Binary classification

A label $X \in \{0, 1\}$ can be modeled using a Bernoulli random variable.

Bernoulli: Shape Changes with p



Binomial Distribution

Definition

If X counts successes in n independent Bernoulli trials with common success probability p,

$$X \sim \mathrm{Bin}(n, p), \qquad P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Key idea: Use the binomial law when the number of trials is fixed and the trials are independent.

Binomial Distribution: Worked Example

Rare outcome in 20 independent trials

Suppose $p = .001$ and $X \sim \text{Bin}(20, .001)$.

$$P(X = 0) = (.999)^{20},$$

$$P(X = 1) = \binom{20}{1}(.001)(.999)^{19}.$$

$$\mathbb{E}[X] = .02, \qquad \text{Var}(X) = .01998.$$

Binomial Probability Mass Function

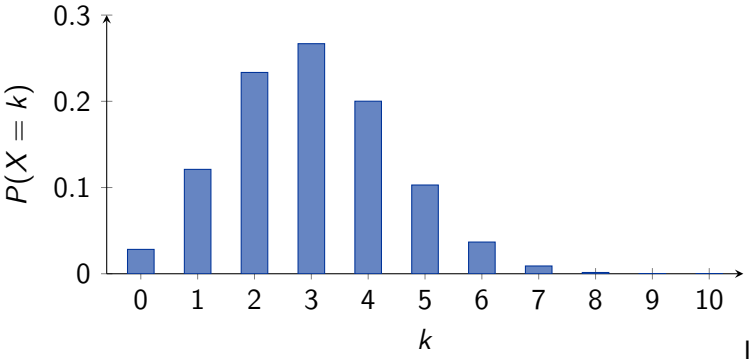


Illustration: $X \sim \text{Bin}(10, .3)$.

Poisson Distribution

Definition

A Poisson random variable models event counts over a fixed interval:

$$X \sim \text{Poisson}(\lambda), \qquad P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!},\ k = 0, 1, \ldots$$

$$\mathbb{E}[X] = \lambda, \qquad \text{Var}(X) = \lambda.$$

Applications

Requests arriving at a server; defects per item; rare event counts.

Poisson Distribution: Worked Example

Average of 3 events per interval

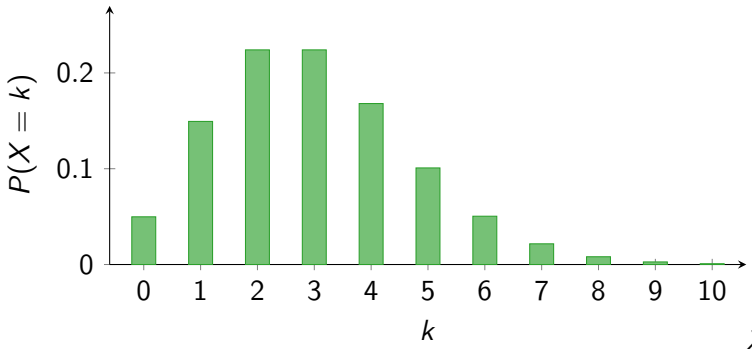
If $X \sim \text{Poisson}(3)$,

$$P(X = 0) = e^{-3}, \quad P(X = 2) = e^{-3}\frac{3^2}{2!}.$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-3}.$$

Key idea: The Poisson law can approximate a binomial law when n is large, p small, and $np \approx \lambda$.

Poisson Distribution: Shape



$X \sim \text{Poisson}(3)$.

Comparing Discrete Distributions

Distribution	Support	Mean	Variance
Uniform $\{1, \dots, n\}$	$1, \dots, n$	$(n+1)/2$	$(n^2-1)/12$
Bernoulli(p)	$0, 1$	p	$p(1-p)$
Binomial(n, p)	$0, \dots, n$	np	$np(1-p)$
Poisson(λ)	$0, 1, \dots$	λ	λ

Key idea: Choose a distribution according to the data-generating experiment, not only the shape of its graph.

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Definition

A continuous random variable has a probability density f satisfying

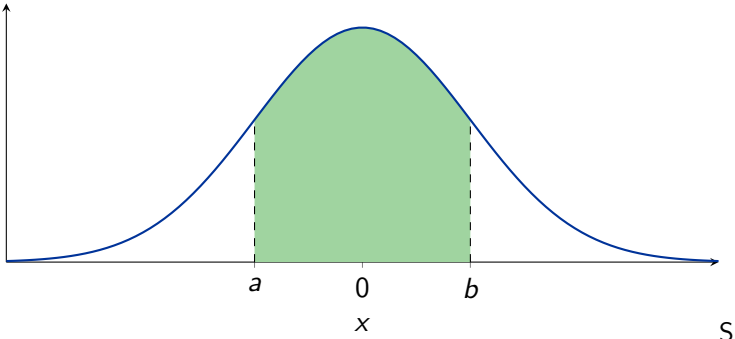
$$f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

For $a < b$,

$$\boxed{P(a \leq X \leq b) = \int_a^b f(x)\,dx}.$$

Key idea: For a continuous variable, $P(X = a) = 0$: probabilities correspond to areas, not individual points.

Probability as Area Under a Density



Shaded area

$$= P(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

Important formulas

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx,$$

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\, dx,$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Cumulative Distribution Function

Definition

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt.$$

If differentiable, $F_X'(x) = f(x)$.

Interval probability

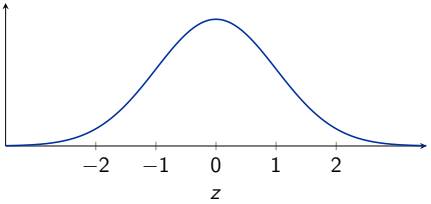
$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Definition

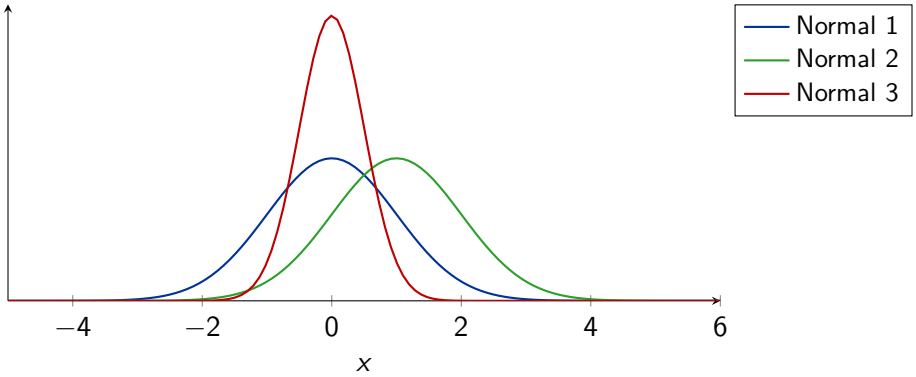
If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$



Effect of Mean and Standard Deviation



Standard Normal Distribution

Standardization

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

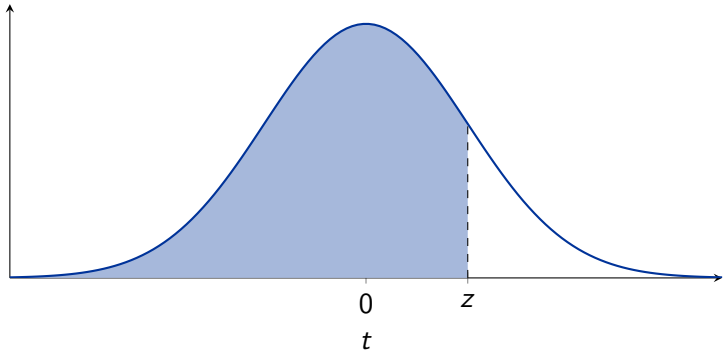
$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Write $\Phi(z) = P(Z \leq z)$.

Interval probability

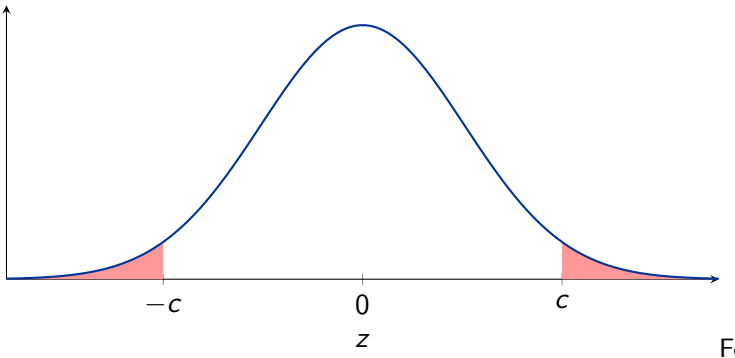
$$P(a \leq X \leq b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

Standard Normal: Left-Tail Probability



$$\Phi(z) = P(Z \leq z).$$

Standard Normal: Two Tails



For $c = 1.96$, the total shaded tail area is approximately 0.05.

Chi-Square Distribution

Definition

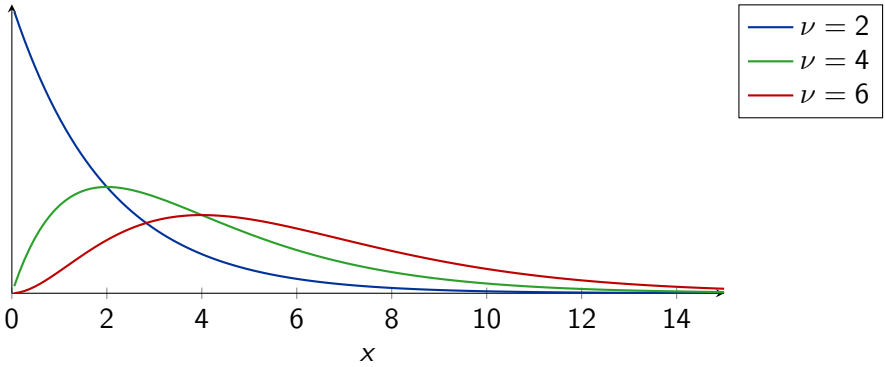
If $Z_1, \ldots, Z_\nu$ are independent standard normal variables, then

$$X = \sum_{i=1}^{\nu} Z_i^2 \sim \chi^2_\nu.$$

$$\mathbb{E}[X] = \nu, \qquad \mathrm{Var}(X) = 2\nu.$$

- Nonnegative, typically right-skewed.
- Important for variance inference and goodness-of-fit tests.

Chi-Square Density for Different Degrees of Freedom



Student's t Distribution

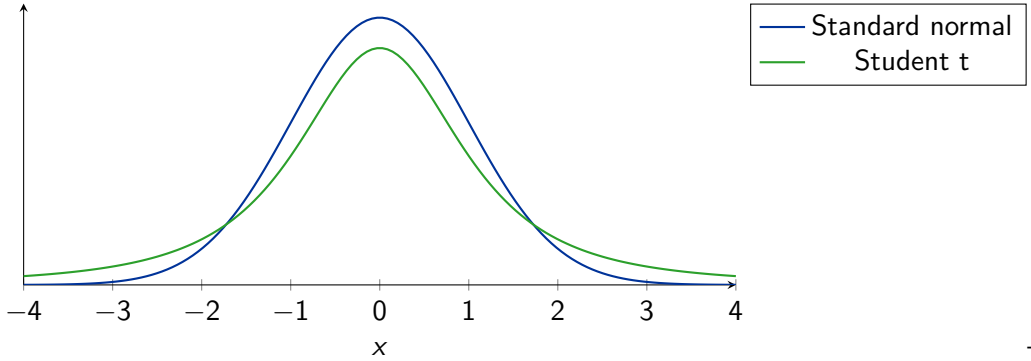
Definition

If $Z \sim \mathcal{N}(0,1)$ and $U \sim \chi^2_\nu$ independently, then

$$T = \frac{Z}{\sqrt{U/\nu}} \sim t_\nu.$$

- Symmetric about zero, with heavier tails than the normal law.
- Approaches $\mathcal{N}(0,1)$ as $\nu \to \infty$.
- Used for inference about means when population variance is unknown.

Student vs Normal: Tail Behavior



The t density has more probability in the tails for small degrees of freedom.

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Central Limit Theorem

Theorem

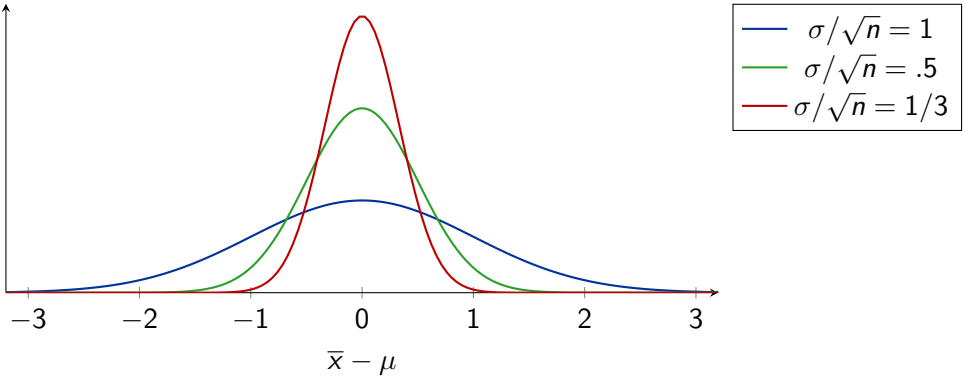
Let $X_1, \ldots, X_n$ be i.i.d. with finite mean μ and variance $\sigma^2 > 0$. Then

$$\boxed{\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \xrightarrow{\mathcal{D}} \mathcal{N}(0,1)}.$$

For large n, $\overline{X}_n$ is approximately $\mathcal{N}(\mu, \sigma^2/n)$.

Key idea: Averages of many independent observations often have an approximately normal sampling distribution.

CLT: Sampling-Mean Distribution Narrows



Normal Approximation of a Binomial

Approximation

For $X \sim \mathrm{Bin}(n, p)$, when np and $n(1-p)$ are sufficiently large,

$$X \approx Y \sim \mathcal{N}\big(np, np(1-p)\big).$$

Standardize with

$$Z = \frac{X - np}{\sqrt{np(1-p)}}.$$

Key idea: Because X is discrete but Y is continuous, apply a continuity correction.

Normal Approximation of a Poisson

Approximation

For a sufficiently large λ,

$$X \sim \text{Poisson}(\lambda) \quad \Longrightarrow \quad X \approx \mathcal{N}(\lambda, \lambda).$$

Example

If $\lambda = 25$, approximate a count by $Y \sim \mathcal{N}(25, 25)$, with standard deviation 5.

Continuity Correction

From discrete counts to a continuous approximation

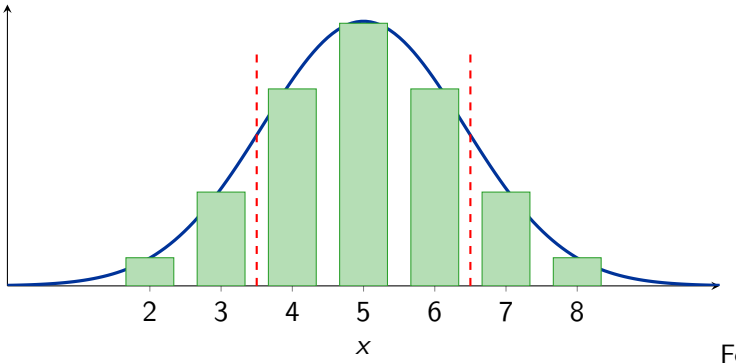
$$P(X \leq k) \approx P(Y \leq k + 0.5),$$

$$P(X \geq k) \approx P(Y \geq k - 0.5),$$

$$P(a \leq X \leq b) \approx P(a - 0.5 \leq Y \leq b + 0.5).$$

Key idea: The half-unit correction matches the boundaries of the discrete bars to continuous areas.

Continuity Correction: Geometric Picture



For $4 \leq X \leq 6$, approximate the probability by the continuous area from 3.5 to 6.5.

Worked Approximation Example

Binomial count

Let $X \sim \text{Bin}(100, .5)$. Then $\mu = 50$ and $\sigma = 5$.

$$P(X \leq 55) \approx P(Y \leq 55.5)$$

$$= P\left(Z \leq \frac{55.5 - 50}{5}\right) = \Phi(1.1).$$

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Probability Models in Machine Learning

Discrete data

- Bernoulli: binary labels.
- Binomial: number of successes.
- Poisson: event counts.

Continuous data

- Normal: measurement variation.
- Student: inference with unknown variance.
- Chi-square: squared normal sums.

Key idea: Bayes' theorem updates beliefs; the CLT connects data averages with statistical inference.

Summary and Revision Questions

1. How do union, intersection and conditioning differ?
2. What distinguishes independent events from disjoint events?
3. When would you use a binomial rather than a Poisson law?
4. How do density, CDF, expectation and variance relate?
5. Why does the continuity correction improve normal approximations?

Thank You!

Questions?

